Solution

SNP-01

Class 08 - Science

1.
   (d) Earthquake of more than 7 magnitude
   Explanation: Earthquake, Tornado, Typhoon, and storm all are destructive but an earthquake of more than 7 magnitudes measured on the Richter scale is most destructive, causing huge losses to life and property.
2. **(a)** Seismic waves
   Explanation: The waves generated on the surface of earth due to sudden movement in a part of earth' s crust are called seismic waves or shock waves. Seismograph is an instrument used to detect and record obtain the intensity of seismic waves generated by an earthquake.
3.
   (c) all of these
   Explanation: all of these
4. **(a)** the crust of the earth
   Explanation: An earthquake is a sudden shaking or trembling of the earth which lasts for a very short time. It is caused by the disturbance deep down inside the uppermost layer of the earth called the crust.
5.
   (c) earthquake
   Explanation: earthquake
6.
   (b) Using a mobile phone
   Explanation: During lightning and thunderstorm we should avoid using telephone, switching electrical switches as all these are connected through wires from where free electrons can easily pass. Mobile phone can be used.
7.
   (c) Tsunami, Floods, Landslide
   Explanation: Earthquake can cause Tsunami, Floods and Landslide. When the lithospheric plates moves, the surface of the earth vibrates. The vibrations can travel all round the earth. These vibrations are called earthquake.
8. **(a)** They occurs suddenly
   Explanation: Natural disaster cannot be predicted in advance because natural disaster occurs suddenly although a warning or predication of its occurrence is nowdays is given with the help of the pictures sent by weather satellites.
9.
   (c) Life and property
   Explanation: Lightening strike could destroy life and property if it falls on building or tree because a flash of lighting carries a lot of electric energy which occurs due to interaction of free flowing ions causing high temperature and which is sufficient to cause fire.
10.
    (b) repel each other
    Explanation: repel each other
11. **(a)** Electric discharge
    Explanation: Losing static electricity as charges move off an object is called electric discharge. Discharged object is electrically neutral and doesn't allow electricity to pass.
12. **(a)** Earthing
    Explanation: The process of transferring a charge from a charged body to the earth is called Earthing.
13.
    (b) Cooking
    Explanation: Earthquake, cyclone and lightning are natural phenomenon as it occurs naturally but cooking is a man-made process.

14.

(c) earthquake under sea

Explanation: earthquake under sea

15.

(d) (i), (ii) and (iii)

Explanation: (i), (ii) and (iii)

16.

(c) Electroscope

Explanation: Electroscope is a device for detecting electric on an object. By using an electroscope, we can tell whether an object is electrically charged or not.

17. **(a)** lightning flashes

Explanation: lightning flashes

18.

(c) Tornado

Explanation: Thunderstorm, cyclone, and earthquake are common natural disaster in India but tornado is common in Japan.

19.

(d) Rise in earth's temperature

Explanation: Carbon dioxide gas is a greenhouse gas. Increase in the level of carbon dioxide gas in the atmosphere leads to greenhouse effect that causes rise in earth's temperature, which leads to global warming.

20.

(c) Air inside expands

Explanation: A candle flame produces heat. When a balloon is taken near candle, the heat causes the air molecules inside the balloon expands that leads to bursting of balloon due to high pressure inside it. The heated air exerts pressure on the walls of the balloon and due to uneven expansion the ballon bursts.

21.

(d) Electron

Explanation: The electron is a subatomic particle that carries a negative electric charge. An electron has a mass that is approximately 1/1836 that of the proton.

22.

(d) Air exerts pressure in direction of sailing

Explanation: It is easier to row a boat in direction of wind because the wind exerts a force on us in the same direction in which our boat is moving and makes our boat to move easily and faster.

23.

(c) Lightning

Explanation: Tsunami refers to the powerful sea waves generated due to the disturbance under the sea. It is not likely to occur in the case of lightning, because lightning is an electric spark caused by the accumulation of charges in clouds.

24.

(b) Crust

Explanation: The outermost layer of earth is called crust. It is that layer which is compartively made of lighter rocks than the mantle and supports life on earth. The middle layer is called mantle and innermost layer is called core.

25.

(d) Benjamin Franklin

Explanation: Benjamin Franklin

26.

(d) Can not be monitored

Explanation: Earthquake activities cannot be monitored as it occurs suddenly without any sign. Satellite can monitor wind and cloud activities only.

27.

(c) It has more chance of falling thunder

Explanation: If a storm is accompanied by lightning, we should not take shelter under a high tree because When a cloud passes over a tall tree, it induces an opposite charge on them. If the charge built is large, it leads to an electrical discharge in the form of a lightning strike, which may shatter the tree or set it on fire causing severe burns and death of the person standing under it.

28. **(a)** Typhoon

Explanation: Cyclones a huge revolving storm caused by very high speed winds blowing around a central area of very low pressure in the atmosphere along with rainfall. A cyclone is called "typhoon" in Japan and Philippines.

29.

(d) Pressure difference between two places is more

Explanation: Moving air is called wind. Wind blows faster when pressure difference between two places is more. Air moves from the regions of high pressure to regions of low pressure in the atmosphere.

30.

(c) Lightening rod at the top of building

Explanation: Building can be protected from lightening by using lightening rod/ conductor at the top of building. Lightening conductor is placed at highest point of the building which can easily attract the free ions passing nearby and transfer it to earth.

31. **(a)** they may attract or repel depending on the type of charges they carry

Explanation: Two charged objects may attract, if they carry opposite charges, while they repel if they carry similar charges.

32.

(c) Vibration of earthquake

Explanation: Seismograph is an instrument used to measure and record the magnitude of an earthquake in terms of the shock waves it produces. It helps in locating the epicentre of earthquake.

33.

(b) Attract each other

Explanation: Like charges repel each other and unlike charge attract each other. This is called the attractive property of a magnet. If the North Pole of a bar magnet is taken close to the South Pole of another bar magnet, it attracts and clings. But if the North Pole of a bar magnet is brought near a the North Pole of another bar magnet it repels or moves away.

34.

(c) Study of climate

Explanation: Flash point, the lowest temperature at which a liquid (usually a petroleum product) will form a vapour in the air near its surface that will “flash,” or briefly ignite, on exposure to an open flame. The flash point is a general indication of the flammability or combustibility of a liquid. Flash point term is related to study of climate.

35.

(b) Wrath of god visiting them

Explanation: In ancient time, people thought that earthquake and lightning g is due to wrath of God visiting them. Later it gets clear that it is due to moving plates of the earth's crust which slide past or collide with one another earthquakes occur and lightning is an electric spark in the sky between oppositely charged clouds.

36.

(d) becomes positively charged while the cloth has a negative charge.

Explanation: When an object is charged by rubbing it against another object, the two objects get oppositely charged. By convention, it is considered that the charged acquired by the glass rod is positive and charged acquired by the cloth is negative. Therefore, the rod becomes positively charged and the cloth becomes negatively charged.

37.

(b) earthquakes

Explanation: When the earth's plates brush past one another or undergo collision, causes an earthquake. An earthquake can cause huge damage to buildings, dams, etc.

38.

(c) Light travels faster than sound

Explanation: Light travels at the speed of 3×10^8 m/s while sound travels at a speed of 332 m/s i.e., light travels much faster than sound that is why lightning is seen earlier and thunder is heard later.

39.

(c) Tornado

Explanation: A tornado is a violent storm with a column of rapidly rotating winds, having the appearance of a dark, funnel-shaped cloud that reaches from the sky to the ground. It causes much damage from the sheer force of its high speed winds.

40. **(a)** Check presence of charge

Explanation: Electroscope is a device used to detect the presence of charge. It consists of two leaves like structure that expand when charged particle passes the charge from the top metallic nob.

41.

(b) Reduced air pressure

Explanation: High speed winds are accompanied by reduced air pressure. Winds blows faster if pressure difference between the two regions is more. Air moves from the regions of high pressure to the regions of low air pressure.

42. **(a)** Loose

Explanation: Loose

43.

(d) Thunder

Explanation: Earthquake may cause floods, landslide and Tsunami but not the thunder because earthquake occurs due to movement of the earth plates whereas thunder is caused due to striking of clouds.

44.

(c) Orissa

Explanation: Orissa is a coastal state in India that is badly affected by cyclone. Bihar, Jharkhand and Karnataka is not affected by cyclone as they are not situated near the coastline but far away from the sea.

45. **(a)** Fault zones

Explanation: The boundaries of earth's plates where earthquake tends to occurs are called fault zone. In this zone there are about a dozen large,irregularly shaped plates which slide over, under and past each other. Movement of these plates occurs in this zone causing earthquakes.

46.

(c) protects the building

Explanation: protects the building

47. **(a)** They are negatively and positively charged and opposite charges attract each other

Explanation: Electron have negative charge and proton have positive charge, opposite charges attract each other, so Electrons and protons attracted each other

48. **(a)** seismic waves

Explanation: seismic waves

49.

(c) In contrast to the attractive force between two objects with opposite charges, two objects that are of like charge will repel each other. That is, a positively charged object will exert a repulsive force upon a second positively charged object. This repulsive force will push the two objects apart.

Explanation: Repel each other

50.

(b) Stand in open field

Explanation: During thunderstorm it is safer to stand in open field as tall tree and building may attract the lightning towards itself because they carry positive charge.

51.

(d) Thunder

Explanation: During lightning, the air gets heated up and expands suddenly. This rapid expansion in the volume of air sends a disturbance through the air producing loud sound. This loud sound is heard as thunder.

52.

(b) Earthquake

Explanation: Earthquake occurs due to movement in tectonic plates of earth crust, volcanic eruptions, or by explosions. Earthquake is not due to pressure change.

53.

(d) Induction

Explanation: Rearranging charges in uncharged metal object without contact with charged object is called induction because it does not required any contact but charge jumps from high end to the lower end.

54.

(c) Wind direction

Explanation: Wind vane is an instrument used to find the direction of wind at a place. It is made of arrows pointing towards different directions and rotates freely. The arrow of the wind vane moves until it points to the direction from which the wind is blowing.

55.

(c) Generate electricity

Explanation: Windmill is used to generate electricity when the vanes of the windmill moves with the speed more than 15km/hr. The energy made by windmills can be used in many ways like it can be used to grind grains,pump water and sawing wood.

56. **(a)** copper wire

Explanation: Two bodies must be joined by any conducting wire, i.e. copper wire for the flow of current.

57.

(d) Electron

Explanation: An object becomes charged when the atoms in the object gain or lose electron. The object that gains electron becomes negatively charged and one who loose electron becomes positively charged.

58.

(c) charged

Explanation: An electroscope is a device, which is used to test whether an object is charged or not.lt consists of closely placed two metallic strips. When both the strips are charged with similar charges, they repel each other and become wide open.

59.

(c) Both may be having either negative or both positive charges.

Explanation: Objects with same charges repel each other i.e. Like charges repel each other. A positive charge repels another positive charge and a negative charge repels another negative charge.

60.

(b) Electric discharge

Explanation: Lightening is a discharge of electricity. A single stroke of lightning can heat the air around it to 30,000°C (54,000°F)! This extreme heating causes the air to expand explosively fast. The expansion creates a shock wave that turns into a booming sound wave, known as thunder.

61.

(c) Seismograph

Explanation: Seismograph

62. **(a)** Seismograph

Explanation: Seismograph is an instrument which detects and records the intensity of seismic waves generated by an earthquake. The graphical record of the intensity of seismic waves is called seismograms.

63.

(c) Earth

Explanation: Earthing is the process of transferring charge from charged object to earth. Earthing of building is done to prevent it from lightening so that free flowing ions can be easily transfer to uncharged.

64.

(c) (i), (ii) and (iii)

Explanation: Earthquake is a sudden shaking or trembling of the earth due to the collision between the earth's plates or brushing past of one plate over the other. It can cause a Tsunami, floods, and landslides.

65. **(a)** -10 °C

Explanation: 14°F = -10°C

66.

(b) Epicentre

Explanation: Epicentre is the point at which a movement occurred which triggered an earthquake. It is the point on earth's surface directly above the focus. Focus is the place inside the earth's crust where the earthquake is generated.

67.

(b) Electron moves out from glass rod

Explanation: When glass rod is rubbed with silk, glass rod becomes positively charged due to movement of electron from glass rod to silk. The materials like glass lose electrons more easily and hence gets positively charged on rubbing . On the other hand materials like silk gain electrons more easily and hence gets negatively charged during rubbing. Whether an object will lose or gain electrons during rubbing depends on the nature of the material of the object.

68. **(a)** Static charge

Explanation: When two insulators are rubbed vigorously over each other, electrical charge is produced. This kind of charge is called as static charge which gets activated due to rubbing.

69.

(b) crust

Explanation: The outermost layer of the earth is carted crust. It is also known as earth's crust.

70.

(b) Increase temperature increase the pressure

Explanation: When a balloon is placed in hot water it gets inflated as an increase in temperature inside the balloon leads to an increase in air pressure inside the balloon and the volume of air on heating and it occupies a bigger space which leads to expansion and the balloon inflates.

71.

(c) Seismic zone

Explanation: Seismic zone is area where chances of occurring earthquake are more than the others. It is the area from where earthquake arises as they are the weak zones of the earth's crust. Since earthquakes are caused by the movement of earth's plates, the boundaries of the plates are the weak zones where earthquakes are most likely to occur.

72.

(d) Install lightning conductor

Explanation: To protect the tall building from the damage of lightning , lightning conductor should be installed at the top of the building so that it can attract free flowing ions and pass them down to the earth. If the lightning strikes,it will hit the top of the lightning conductor rather than the building and gets discharged safely into the ground through the buried metal plate.

73.

(c) (i) and (ii)

Explanation: Since earthquakes are caused by the movement of plates, the boundaries of the plates are weak zones where earthquakes are more likely to occur. These weak zones are also known as seismic or fault zones.

74.

(c) A copper rod

Explanation: Only non-conducting materials can be easily charged by friction. Copper is a highly conducting material. Therefore, a copper rod cannot be charged easily by friction.

75.

(c) Earthquake

Explanation: The richter scale is scale of numbers used to tell the power (or magnitude) of earthquakes. The richter scale calculates an earthquake's magnitude from the amplitude of the earthquake's largest seismic wave recorded by a seismograph.